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Week 5 Feasibility Memo Outline
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AI-Assisted Triage: Data Exploration

One-sentence verdict

The ED triage dataset appears good enough to build a first basic triage model but important limits must be handled carefully, especially very few ESI 1 (most urgent) cases, sparse chief complaint flags, some columns that could leak future information into the model and the need to validate the model before any clinical use.

Dataset summary

- Dataset: `yaleemmlc_admissionprediction_triage.csv`
- The dataset contains 55,121 emergency department arrival records and 225 columns.
- The main target variable is ESI (Emergency Severity Index), the triage level assigned to each patient. ESI runs from 1 (most urgent) to 5 (least urgent).
- The data includes demographics (age, gender, race, ethnicity), arrival and triage details, vital signs (heart rate, blood pressure, respiratory rate, oxygen saturation, temperature, glucose) and 200 binary chief complaint columns (each starting with cc_).
- Most patients fall in the middle acuity levels: ESI 3 is the largest group at 27,010 patients (49.0%), followed by ESI 2 at 17,924 patients (32.5%). ESI 1 is very rare with just 77 patients (0.1%), meaning there are few examples for a model to learn from when it comes to the sickest patients.
- There are 0 missing ESI labels, so every patient has a recorded triage outcome to train against.

Top 3 data-quality concerns

1. ESI class imbalance

Only 77 patients (0.1%) are ESI 1. A model could look accurate overall while still doing a poor job spotting the sickest patients, since it has so few examples to learn from. ***Planned response:*** In Week 6, check model performance separately for each ESI group, especially ESI 1 and ESI 2, so weak performance on the most urgent cases isn't hidden by strong overall accuracy.

2. Sparse chief complaint features

Of the 200 chief complaint columns, 149 appear in less than 0.5% of encounters. These rare columns are unreliable as standalone predictors, there isn't enough data behind them to trust the patterns. ***Planned response:*** Group, exclude, or treat rare chief complaint features cautiously during modelling.

3. Possible leakage columns

Columns such as ‘disposition’ and ‘previousdispo’ may reflect information that wasn't actually available at the time of triage. A real triage model can only use what a nurse would know at the time of assessment.

Planned response: Exclude leakage-prone columns from the model inputs and treat the Week 6 model as an early baseline only, not something ready for clinical use.

Top 3 reasons to proceed

1. There's a real triage outcome to learn from

With 0 missing ESI labels, the model can be trained on the actual triage decision rather than a stand-in for it.

2. Key triage features are already there

The dataset includes clinically meaningful information available early in a visit such as vital signs, demographics, arrival details and chief complaints.

3. Early patterns line up with clinical expectations

Some early findings match what a clinician would expect. For example, chest pain skews toward more urgent ESI levels and oxygen saturation shows the strongest relationship with ESI of any vital sign.

Planned visual evidence

The final memo will draw on the Week 5 data-quality dashboard, including:

- Missingness plot
- ESI and age distribution plot
- Race and ethnicity distribution plot
- Chief complaint frequency plot
- Vitals by ESI plot
- Vitals-and-ESI correlation heatmap
- Equity plot showing ESI distribution within each race category

Top-10 feature shortlist for the final memo

A ranked shortlist of features to test in Week 6:

1. triage_vital_o2 - Low oxygen levels are a warning sign for breathing problems and had the closest link to ESI of any vital sign.
2. triage_vital_rr - A faster breathing rate can be a sign of distress, infection, pain, or a lung problem.
3. triage_vital_hr - A higher heart rate can point to shock, pain, fever, anxiety, dehydration, or sudden illness.
4. triage_glucose - Abnormal blood sugar can signal that the body is unstable and may affect how urgent the case is.

5. cc_chestpain - Chest pain is a classic warning sign in the ED and was linked to more urgent cases.
6. cc_shortnessofbreath - Trouble breathing is urgent and was linked to more urgent cases.
7. cc_suicidal - Thoughts of suicide often need urgent safety checks and were linked to more urgent cases.
8. cc_alteredmentalstatus - Sudden confusion or unresponsiveness is a major warning sign of serious illness or a brain-related problem.
9. cc_syncope - Fainting can point to a heart, nerve, or blood-flow problem and may need urgent assessment.
10. cc_abdominalpain - Abdominal pain is common in the ED and can range from minor illness to serious surgical or infection-related problems.

Caveats for the final memo

- Correlation shows a relationship, not proof that a feature causes higher or lower acuity.
- The Week 6 model is a first baseline, not something ready for hospital use.
- ESI 1 is rare, so performance on the sickest patients needs close, separate checking.
- Many chief complaint columns are rare and may need to be grouped before modelling.
- Temperature appears to be recorded in Fahrenheit, so fever/low-temperature cutoffs should use Fahrenheit units.
- Disposition-related variables should be excluded to avoid leakage.
- The model should be tested in the exact setting where it will be used. One ED's data may not represent every Caribbean hospital or patient population.